

Industry Analysis

2024



TELECOMMUNICATIONS



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Industry Definition

Telecommunication Defined

According to The Telecommunication Act, 2023 - "telecommunication" means transmission, emission or reception of any messages, by wire, radio, optical or other electro-magnetic systems, whether or not such messages have been subjected to rearrangement, computation or other processes by any means in the course of their transmission, emission or reception;

Industry Definition

The telecommunication industry is essential to economic growth, supporting sectors like e-commerce, finance, IT, healthcare, and logistics by enabling global data transfer. Constantly evolving with new technology, telecom drives productivity and has become integral to daily life. In India, it divides into three main segments: **Telecom Service Providers, Telecom Equipment Manufacturers, and Telecom Infrastructure.**

This report presents the telecommunication industry to you.

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Telecommunication drives economic growth and is essential to modern life.

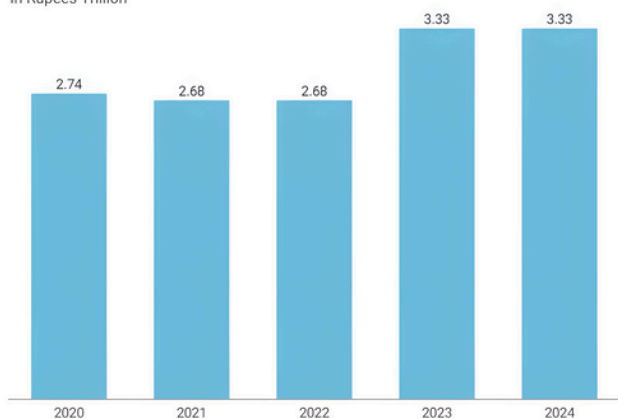


Size of the Industry

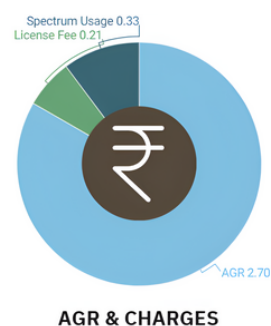
The size of the telecom sector is analyzed through metrics such as **gross revenue** and segment-wise **subscribers** over the past five years.

Gross Revenue

In Rupees Trillion



Source : TRAI



AGR & CHARGES

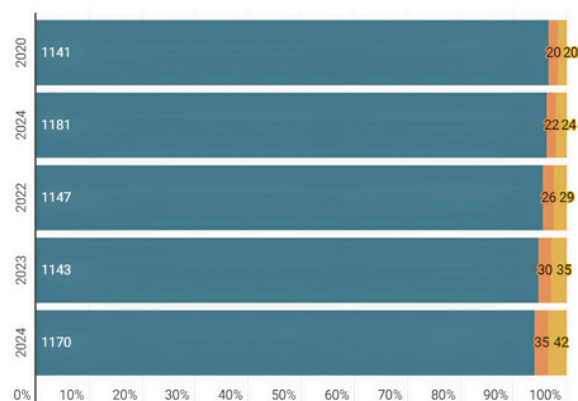
1,247 MILLION

SECOND-LARGEST SUBSCRIBER BASE IN THE WORLD

Subscriber Base

In Millions

● Mobile Wireless ● Wireline ● Internet Wired



Source : TRAI

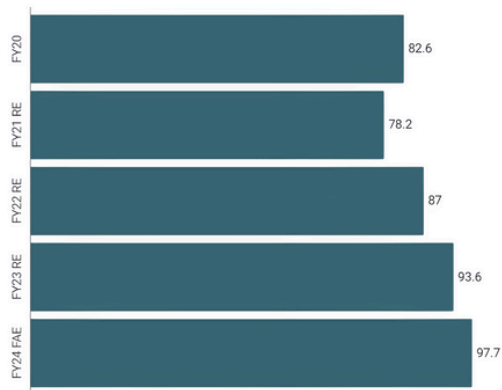
The **gross revenue** of the telecom sector was **₹3.36 lakh crore in 2024**, which grew at a **CAGR** of approximately **4.16%** over the last five years. Mobile wireless subscribers, the most commonly used service, have remained stagnant, while wireline subscribers have shown steady growth over the last five years. Additionally, internet wired (broadband) subscribers have increased steadily alongside wireline subscriptions, contributing to a total of **1,247 million subscribers**.

Key Demand Drivers

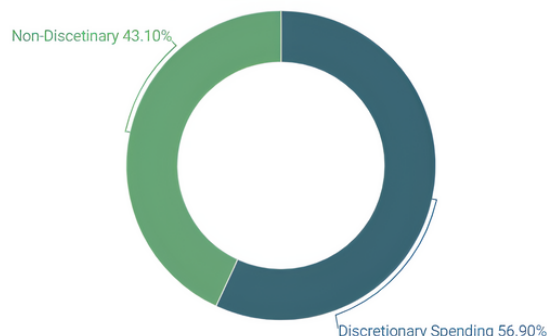
a. Consumption expenditure largest contributor to GDP

PFCE

In Rs Trillions



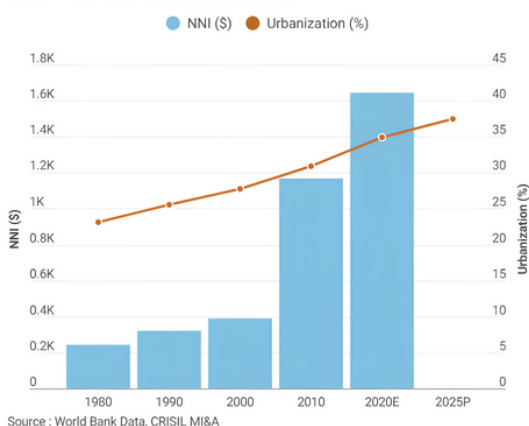
RE: Revised Estimates, PE: Provisional Estimates, FAE: First Advance Estimates
Source: MoSPI, CRISIL MI&A



- PFCE (Private Final Consumption Expenditure) is the largest component of India's GDP, contributing 56.9% in the current fiscal.
- Between FY20 and FY24, PFCE is projected to grow at a 4.29% CAGR.
- Increasing discretionary spending since FY20 indicates higher disposable income and spending capacity.

b. Urbanization and rise in income will drive long-term growth

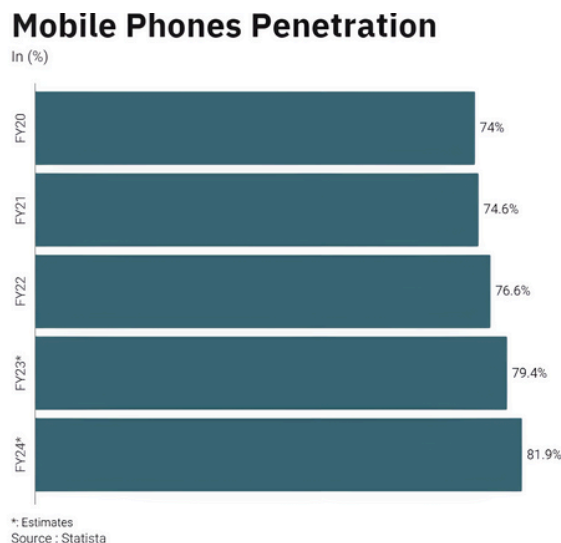
NNI & Urbanization



Per Capita National Net Income grew at a **CAGR** of **4.89%** from 1980 to 2020.

- Urbanization is a key growth driver for India's economy.
- Rising urbanization to boost infrastructure investments, job creation, modern services, and consumption/saving potential.

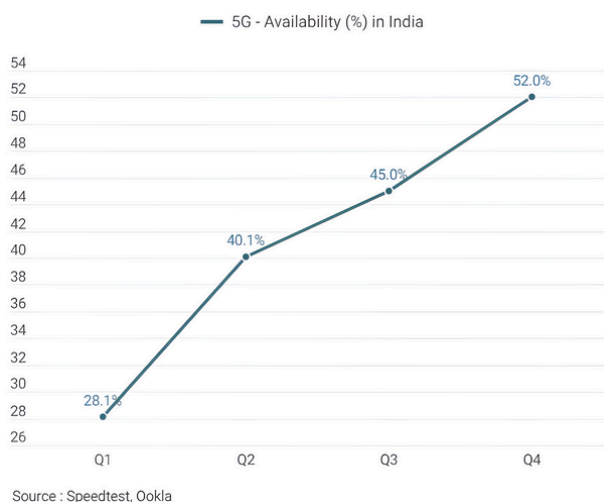
c. Mobile phone penetration to drive growth in the telecom sector



- Mobile Phones Penetration in India has increased from 74% in FY20 to 81.9% in FY24.

- This rise in mobile penetration is expected to stimulate further demand for telecom services.

d. 5G Rollout in India



India is **one of the fastest** countries to roll out 5G globally

- Ookla's Speedtest Intelligence data indicates a rise in 5G Availability in India from 28.1% in Q1 2023 to 52.0% in Q4 2023, reflecting a 23.9 percentage point increase over the year.

- The surge in 5G-capable smartphones is a key factor in the increase of 5G Availability

Competitive Intensity

- The telecom industry is **capital-intensive and highly competitive**, making M&As crucial for survival and tech advancements.
- M&As often involve horizontal integration, expanding customer base and diversifying products.
- Fewer market players enable economies of scale, reducing infrastructure overlap and operational costs, with **2-5 companies** typically dominating major markets.

Countries		Top Telecom Companies
Australia		Telstra, TPG, Singtel Optus, Macquarie
Brazil		Telefonica, Claro, TIM
China		China Mobile, China Telecom, China Unicom
France		Orange, Free Mobile, SFR, Bouygues
India		Reliance Jio, Bharti Airtel, Vodafone Idea, BSNL
Spain		Vodafone Spain, Orange Spain, MasMovil, Telefonica
United States		Verizon, T-Mobile, AT&T

Source : Research, Internet

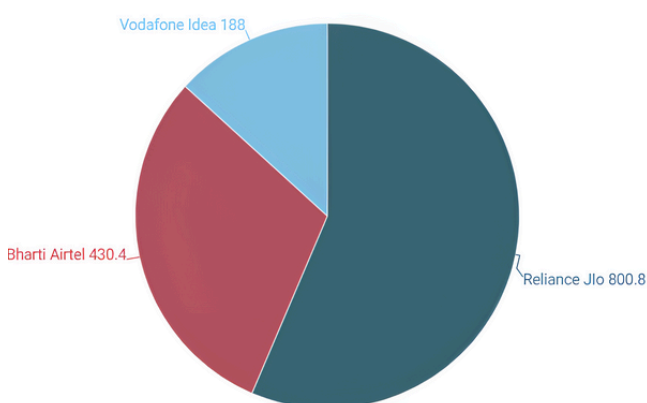
a. High capital investments vital for sustaining telecom services

- Telecom industry requires significant capital for spectrum purchases and infrastructure maintenance.
- Rs 1,500 billion spent in the 5G auction (2022), with Reliance Jio, Bharti Airtel, and Vodafone Idea spending ₹800.8B, ₹430.4B, and ₹188.0B, respectively.
- Indian telecom operators spent ~₹3,000 billion on spectrum acquisition since 2014.

- Industry's low average asset turnover ratio of 0.3 (FY20-23) indicates high sustained capex needs.
- Capex intensity was ~30% of revenue in FY23, projected at ~25% in FY24 and FY25.
- Network opex accounted for 25-30% of revenue (FY21-23).

5G Auction (2022)

In Rs Billion



0.30

Asset Turnover Ratio



30%

Capex Intensity



25-30%

Network Opex

b. Hurdles in establishing network coverage across India and competing with established players

- India's vast and rugged terrain makes telecom infrastructure setup complex and costly.
- New players face challenges in matching established companies' coverage.
- Cost per customer is higher in rural areas due to lower population density.

c. Economies of scale important for survival and profitability of players

- Low ARPU in India makes achieving economies of scale crucial for telecom sector survival.
- High capital costs and network deployment expenses deter new entrants without a large customer base to ensure returns.

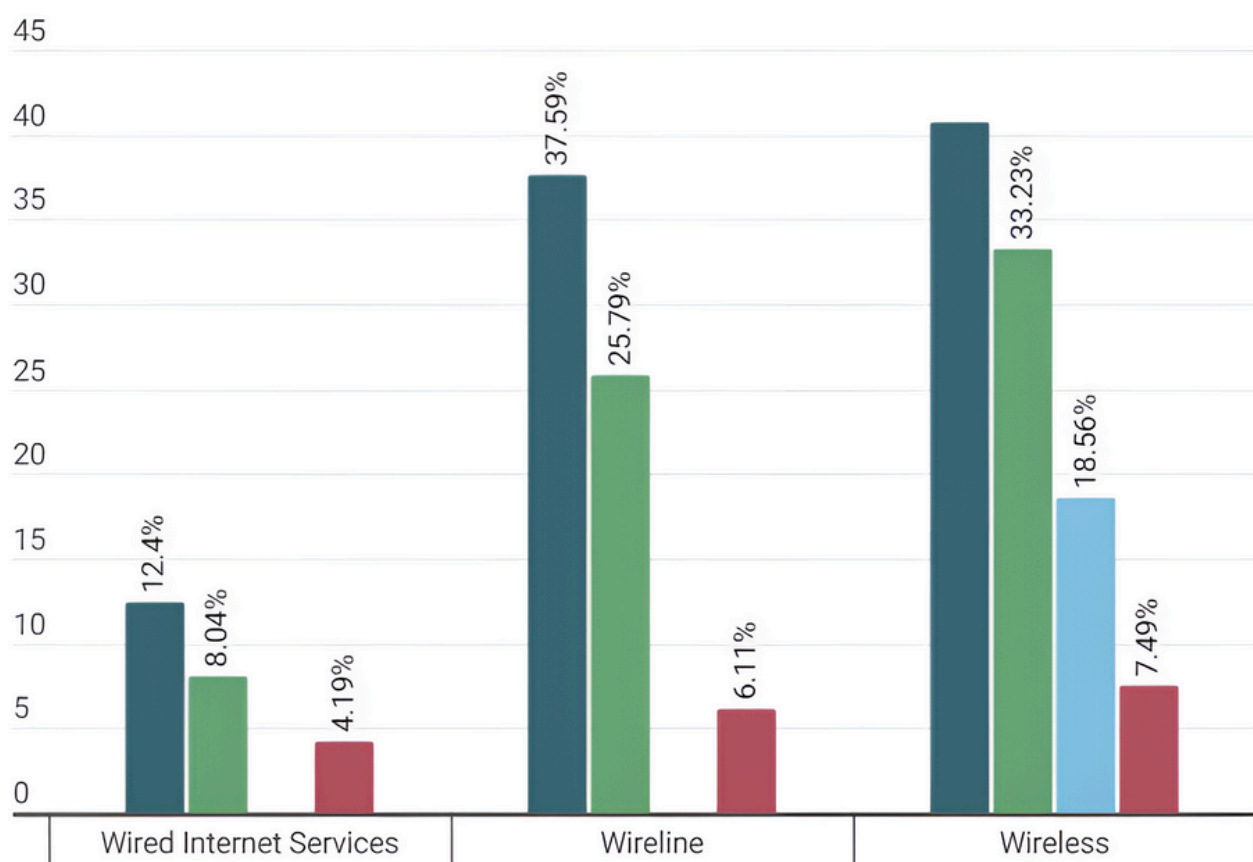
Key Players



Market Share (2024)

In (%)

● Reliance Jio ● Bharti Airtel ● Vodafone Idea ● BSNL / MTNL



Source : TRAI

Key Sectoral Risks

Financial Stress	AGR Dispute
- Telecom industry debt hit ~₹6.4 lakh crore by March 31, 2023 (ICRA). - High spectrum costs, tariff wars, and infrastructure investments drive financial stress. - Vodafone Idea's government dues total ₹2.1 lakh crore. - Financial strain has led to reduced capex and delayed 5G rollouts. - Some companies have exited the market due to these pressures.	- AGR remains a major challenge for Indian telecom operators. - The 2019 Supreme Court ruling expanded AGR to non-telecom revenues, adding ₹1.69 lakh crore in liabilities. - Government offered a moratorium and equity conversion, but balance sheets are still strained. - Financial strain limits investment in new technologies and infrastructure.

Infrastructure Gaps

- India's telecom infrastructure shows a significant **urban-rural divide** as of March 2023.
- Urban **tele-density was 133.81%**, while **rural** tele-density was only **57.71%**.
- Challenges in rural areas include difficult terrain, inconsistent power supply, and lower return on investment.

Spectrum Pricing

- In the 2022 5G spectrum auction, **the government earned ₹1.5 lakh crore**, but operators claim high costs hinder network expansion and quality.
- India's **spectrum prices are among the highest globally**, affecting telecom companies' financial health.
- This situation **may slow** the adoption of new technologies like **5G**, impacting India's digital transformation.

Porter's Five Forces Analysis

